

Retrieval Architecture

The retrieval layer combines two independent signals. Dense vectors capture semantic similarity, so a question phrased differently from the source text still matches. Sparse BM25 vectors capture exact terminology, so rare identifiers and proper nouns are not lost in the embedding. The two ranked lists are combined with reciprocal rank fusion inside the vector database. Fusion happens server side because moving both candidate lists across the network only to re-sort them wastes a round trip.

Chunking Policy

Chunks target eight hundred characters with one hundred and fifty characters of overlap, and splits land on sentence boundaries. Page boundaries are never crossed: a chunk that spans two pages has no single page number, and page attribution is the guarantee the system makes. Transcript segments are the opposite case and are merged, because a single whisper segment is a few seconds of speech and too small to retrieve against. Merging widens the timestamp span, which loses nothing.

Abstention Policy

The sufficiency gate is deterministic. It compares the number of chunks scoring above a configured threshold against a configured minimum. When the corpus does not clear the bar after three iterations, the system abstains instead of answering. Abstention is the correct output when evidence is thin, not a failure of the system. The thresholds are configuration rather than constants so that the evaluation can sweep them.